PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo: PRISM

Titolo completo: Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements

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Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

1 State of the art (4000 caratteri)

A central challenge in materials characterization is determining how electronic and local atomic structure evolve under realistic conditions. Changes induced by irradiation, processing, phase transformations, chemical reactions or device operation may be transient, reversible or environment-dependent, and can therefore be missed by ex-situ measurements [1]. This requires in-situ and/or operando techniques capable of probing thick, heterogeneous or diluted samples while retaining sensitivity to selected elements and their local coordination [2,3]. Hence, there is a broad need for non-destructive, element-selective and bulk-sensitive probes compatible with complex environments and long acquisition campaigns.

X-rays are particularly suited for such investigations because of their high penetration depth with respect to other optical techniques, enabling access to the bulk of samples. By tuning the photon energy of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. These properties make X-ray methods well-suited to non-destructive in situ and operando studies [4].

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, the XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge, the EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. XAS can be measured in transmission, by recording the attenuation of the incident beam through the sample, or in fluorescence, by detecting the characteristic X-rays emitted after absorption. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, broad energy tunability and dedicated beamlines enable high-quality spectra with excellent signal-to-noise ratio over a wide energy range. X-ray free-electron lasers further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7].

These limitations have driven renewed interest in laboratory XAS systems based on conventional X-ray tubes [8]. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [9] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [10].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high

hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.
As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The **main goal** of the project is **to develop and validate a room-temperature laboratory platform for fluorescence-mode X-ray absorption spectroscopy (f-XAS) above 20 keV**. Source, crystal optics, sample geometry, CZT detection, electronics and simulations will be optimized as a single measurement chain to obtain chemically meaningful XANES spectra from samples for which transmission is impractical.

The project is organized into the following specific **goals (G)** and **expected outcomes (EO)**:

G1 – Establish a validated design and predictive model of the complete instrument.
EO1 – An integrated simulation workflow and a frozen experimental design, validated at the Ag K-edge and at one energy of at least 40 keV. Predictions and measurements will agree within 20% for detected rate and within 10% for spot size and energy resolution.

G2 – Develop and characterize a multichannel CZT detector operating at room temperature.
EO2 – A calibrated system with intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better over 20–40 keV, dead time below 10% and residual pile-up below 5% at the selected operating rate.

G3 – Validate the complete platform on reference compounds with different chemical states.
EO3 – Quantitative Ag K-edge XANES with effective XAS resolution of 10 eV FWHM or better, edge-position reproducibility within 1 eV, edge-step SNR of at least 20 and acquisition time below the 12 h laboratory benchmark. The normalized spectral deviation from reference data will be below 5%.

G4 – Demonstrate f-XAS on at least one thick, diluted or operando-compatible sample.
EO4 – Discrimination of at least two chemical states through an edge shift of at least 1 eV and/or a normalized white-line variation of at least 5%, with a significance of at least 3σ and an acquisition time below 12 h per spectrum. The measured operating domain will be reported in terms of concentration, sample geometry and acquisition time.

3 Methodology (10000 caratteri)

PRISM will follow a requirements-driven and simulation-informed development strategy. The experimental requirements will first be translated into quantitative specifications for the X-ray source, crystal optics, sample geometry, CZT detector and DAQ. End-to-end simulations will then guide the design and co-optimization of these subsystems before procurement and construction. The resulting instrument will be assembled together with the custom multichannel CZT detection system and validated through a staged programme, progressing from subsystem characterization and calibration to measurements on reference compounds and, finally, realistic samples. Experimental results will be continuously compared with the simulations, establishing a closed simulation–measurement loop that will support design refinement, performance verification and reliable interpretation of the measured spectra.

Requirements, simulations and design freeze

An end-to-end, source-to-detector simulation of the spectrometer will be developed to define the experimental requirements and freeze the main design parameters before procurement and construction. The Ag K-edge at approximately 25.5 keV will be used as the reference case for defining the XANES measurement protocol and the required performance in terms of energy resolution, photon flux, signal-to-background ratio and acquisition time [11].

The simulation workflow will consist of two coupled stages. First, SHADOW4 ray-tracing simulations [12] building on the approach previously validated for the VOXES spectrometer [13], will model photon transport from the X-ray tube through the collimating slits and crystal optics to the sample. Different silicon- and germanium-based crystal reflections will be investigated in the VOXES-type von Hamos geometry. Parametric studies will optimize the crystal material, reflection, curvature radius ρc\rho_cρc, dimensions, source–crystal and crystal–sample distances, slit apertures and Bragg geometry. The simulations will quantify the geometrical acceptance, energy bandwidth, photon flux, beam profile and illuminated spot size at the sample, while also evaluating alignment and mechanical tolerances.
The SHADOW4 output will then be transferred to GEANT4, which will simulate photon absorption in the sample, fluorescence emission, self-absorption, elastic and inelastic scattering, background contributions and energy deposition in the multichannel CZT detector. The combined simulations will be used to co-optimize the source, crystal optics, slit system, sample geometry and detector configuration, balancing count rate, spectral resolution and background rejection.
A coordinated motorized scanning scheme will also be defined to vary the incident energy across the Ag K-edge while maintaining the required source–crystal–sample geometry. The final outcome of this activity will be a design freeze specifying the X-ray source, crystal type and dimensions, curvature radius, distances, slit apertures, sample position, CZT geometry, motion stages and required positioning accuracy.

CZT/DAQ development and instrument integration

To measure the fluorescence spectrum emitted by the sample at energies above 20 keV, a custom multichannel cadmium-zinc-telluride (CZT) detector system will be designed and implemented. Unlike general-purpose commercial detectors, its geometry, segmentation and number of channels will be defined from the photon-flux, background and count-rate requirements established in WP1. A baseline crystal thickness of 2 mm will provide high detection efficiency in the target energy range while maintaining compactness and room-temperature operation; the final thickness and active area will be confirmed through GEANT4 simulations. Sensitivity at lower energies will also be retained to identify and reject background components or contamination arising from lower-order crystal reflections.

Dedicated low-noise front-end electronics and a multichannel data-acquisition system will be developed to provide synchronized waveform acquisition, stable triggering and channel-by-channel calibration. Pulse-reconstruction and equalization algorithms will correct differences in gain, baseline and response among channels. Pulse-shape-analysis techniques, building on methods previously developed for germanium detectors, will be used to reject pile-up and distorted events and to mitigate the effects of incomplete charge collection and charge trapping in the CZT crystals. Dead-time and pile-up corrections will be implemented to preserve spectral accuracy over the expected range of incident rates.

The detector system will be characterized in terms of absolute and relative efficiency, energy resolution expressed as FWHM, gain stability, linearity and maximum useful count rate. These measurements will define the operating point that provides the best compromise between efficiency, spectral resolution and acquisition time. Finally, the CZT array, front-end electronics and DAQ will be integrated with the detector mechanics, motor-control system and instrument input/output monitoring, enabling coordinated acquisition with the monochromator energy scan and continuous recording of the relevant experimental parameters. This activity forms part of the broader requirements-driven development of the PRISM fluorescence-XAS instrument.

Commissioning and quantitative reference validation

Following assembly and integration, the instrument will undergo a structured commissioning and quantitative validation programme. The multichannel CZT detector will first be calibrated channel by channel to determine the energy scale, gain, offset, linearity and energy resolution. In parallel, the incident-energy scale of the monochromator will be established using reference absorption features and verified across the scan range. Dedicated measurements will characterize the beam spot at the sample, energy bandwidth, photon rate, spectral background and stability, providing a direct comparison with the SHADOW4 and GEANT4 predictions. The detector response will also be tested at energies extending to at least 40 keV to verify the performance of the system beyond the Ag K-edge.

The complete XAS measurement chain will then be validated at the Ag K-edge. Repeated XANES scans of a metallic Ag foil will be used to assess energy calibration, scan-to-scan reproducibility, signal-to-noise ratio and acquisition time. At least two additional Ag compounds representing different oxidation states or local chemical environments will subsequently be measured. Their edge positions and XANES features will be compared with reference spectra to verify that the instrument can distinguish chemically meaningful differences. The final validation will quantify the achieved energy resolution, background rejection, reproducibility, stability and useful count rate, and will determine whether the performance requirements defined during the design phase have been met.

Realistic-sample demonstration

After validation on reference materials, the capability of PRISM will be demonstrated on a realistic Ag-containing sample selected to represent conditions in which fluorescence detection is advantageous, such as a thick, diluted, heterogeneous or operando-compatible specimen. Two controlled and reproducible chemical states will be prepared to produce measurable differences in the Ag local electronic environment while keeping the sample composition and geometry sufficiently well characterized for quantitative comparison.

Before the measurements, GEANT4 simulations and analytical attenuation models will be used to estimate incident-beam absorption, matrix effects and self-absorption of the emitted fluorescence. Repeated fluorescence-XAS scans will then be acquired under identical conditions to evaluate reproducibility and sensitivity to chemical changes. The spectra will be analysed in terms of absorption-edge position, white-line intensity and other relevant XANES features, and the statistical significance of the observed differences will be quantified.
Measurements and simulations will finally be repeated for selected Ag concentrations, sample thicknesses, geometries and acquisition times. This study will define the practical operating limits of the instrument and establish the conditions under which reliable chemical-state discrimination can be achieved on realistic samples. The outcome will therefore be both a proof-of-capability measurement and a quantitative performance map linking sample properties and measurement geometry to signal-to-noise ratio, spectral distortion and required acquisition time.

4 Organisation (4000 caratteri)

Name	Unit	FTE
Simone Manti	INFN, Section of Frascati	1.0
Alessandro Scordo	INFN, Section of Frascati	0.3
Catalina Curceanu	INFN, Section of Frascati	0.1

WP1-Requirements, simulations and design freeze

1. d
2. d

	Year I				Year II			
	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WORK PACKAGES (WP)								
WP1: Simulation and Optimization	■	■	■					
T1.1 - Requirements definition	■							
T1.2 - SHADOW4 ray-tracing optimization	■	■						
T1.3 - GEANT4 simulations		■	■					
T1.4 - Final geometry selection			■					
WP2: Instrument design and procurement		■	■	■	■			
T2.1 - Mechanical design		■	■					
T2.2 - Procurement of optics, mechanics and detector			■	■				
T2.3 - Integration				■	■			
T2.4 - Commissioning					■			
WP3: Assembly and validation at VOXES				■	■	■		
T3.1 - Energy calibration				■	■			
T3.2 - Resolution and efficiency measurements					■	■		
T3.3 - XAS/XES validation with laboratory sources					■	■		
T3.4 - Readiness review						■		
WP4: Experimental Demonstration						■	■	■
T4.1 - Installation at EuAPS						■		
T4.2 - First XAS measurements						■	■	
T4.3 - Feasibility demonstration							■	■
T4.4 - Final data analysis								■
MILESTONES (MS)								
MS1. Optimized spectrometer design completed		x						
MS2. Hardware assembled and commissioned				x				
MS3. Laboratory validation completed						x		
MS4. Successful demonstration at EuAPS								x

5 Synergies with Other Activities (3000 caratteri)

EuAPS
VOXES
PANDORA

6 Impact and Possible Spillovers (3000 caratteri)

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7 Costs

-X-ray microfocus tube
-Crystals
-Axis
-Sample targets

-EXRS2028 Paris?

Year			

CZT			

8 Risk Assessment

9 References

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